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Canadian National Project Proposal

AI-IDP — Universal AI Identity, Delegation, Provenance, Traceability, Quality, Accountability, and Permanent Audit Standard

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1. Cover

This is a proposal to the Government of Canada for the adoption of AI-IDP as a National Standard of Canada, the integration of AI-IDP registration into the Treasury Board Directive on Automated Decision-Making, the establishment of a national AI-IDP registry authority, and the enactment of supporting legislation (the proposed AI Actor Identity and Traceability Act). The proposal is grounded in current Canadian law (PIPEDA, Privacy Act, provincial privacy statutes, Constitution Acts 1867 and 1982, Treasury Board Directive on Automated Decision-Making) and references proposed legislation (AIDA under Bill C-27) as proposed, not enacted. The proposal distinguishes current law from proposed law throughout.

2. Document Control

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3. Author and Organization

Pierre-Edward Procyk is the project originator, principal conceptual author, framework architect, principal research author, owner of the original integrated proposal, and final authority for external publication, licensing, submission, and release. Cognitive Industries — Les Industries Cognitives is the proposing organization. Brand positioning: Premium Cognitive Systems. Location: Saguenay, Québec, Canada.

4. Executive Summary

Canada faces an AI accountability gap. AI agents now make material decisions across Canadian society — in customer service, software development, financial analysis, healthcare support, education, public-sector administration, and operations. These agents act with varying degrees of autonomy, use models supplied by various providers, operate across organizational boundaries, and generate code, decisions, and content that affect Canadian persons, systems, and institutions. Current accountability mechanisms — application logs, audit trails, Git history, cloud-provider audit logs — are fragmented, provider-specific, and insufficient for cross-organization incident reconstruction. The accountability gap creates harm: organizations cannot reliably answer "which agent did what, when, on whose authority, using which model and provider" after an incident, and affected persons cannot reliably attribute harm to a specific accountable party.

AI-IDP is a proposed universal Canadian framework that addresses this gap. AI-IDP requires every operational AI agent used, created, deployed, distributed, controlled, executed, or made available in Canada to possess a unique persistent AI Actor Identifier, runtime instance identifiers, traceable relationships to provider/model/deployment/controller/principal/parent/child/task/action/tool/resource, a permanent append-only tamper-evident history, quality evidence for AI-created or AI-modified code and systems, a tiered registry model, independent auditability, incident-reconstruction capability, legal accountability, and conformity evidence. The framework's central normative principle is "no valid AI actor identity, no lawful agent operation."

This proposal recommends: (1) adoption of AI-IDP as a National Standard of Canada through the Standards Council of Canada; (2) integration of AI-IDP registration into the Treasury Board Directive on Automated Decision-Making; (3) establishment of a national AI-IDP registry authority with provincial federation agreements; (4) establishment of tiered conformance levels (L1–L4) with sectoral certification bodies; (5) establishment of small-developer and open-source compliance profiles; (6) establishment of enforcement options; (7) enactment of the proposed AI Actor Identity and Traceability Act; (8) public consultation; (9) parliamentary committee review; (10) annual regulator reporting to Parliament.

5. National Problem

The national problem is the AI accountability gap. Specifically: (1) there is no universal persistent identifier for AI agents operating in Canada; (2) there is no universal record of AI agent actions and their authority chains; (3) there is no universal tamper-evident history that survives agent termination, provider closure, model retirement, repository transfer, or organizational restructuring; (4) there is no universal quality-evidence requirement for AI-generated code and systems; (5) there is no universal registry model; (6) there is no universal incident-reconstruction capability; (7) there is no universal conformity and certification framework. This gap creates harm to affected persons, organizations, regulators, and insurers. The gap is widening as AI deployment grows.

6. Current Canadian Context

6.1 Current Law and Policy

The current Canadian legal and policy context includes: PIPEDA (private-sector privacy); the Privacy Act (public-sector privacy); the Constitution Acts 1867 and 1982 (including the Canadian Charter of Rights and Freedoms); the Criminal Code (computer-crime, fraud, identity-related offences); the Copyright Act (authorship and AI-generated works); provincial privacy statutes (Quebec Law 25, BC PIPA, Alberta PIPA); provincial freedom-of-information and privacy acts; provincial consumer-protection statutes; the Treasury Board Directive on Automated Decision-Making; the Algorithmic Impact Assessment (AIA) instrument; Canadian Centre for Cyber Security guidance; Communications Security Establishment guidance; Office of the Privacy Commissioner of Canada guidance; provincial privacy regulators' guidance; Truth and Reconciliation Commission of Canada: Calls to Action; First Nations Information Governance Centre: OCAP® principles; Inuit Tapiriit Kanatami and Métis National Council policy positions; CanLII case law.

6.2 Proposed Legislation

The proposed Artificial Intelligence and Data Act (AIDA), Bill C-27, is the most recent federal horizontal AI bill of record. AI-IDP references AIDA as proposed legislation (not enacted). If AIDA is enacted, replaced, or withdrawn, the legal analysis and government proposal must be updated to reflect the new state.

6.3 Provincial Developments

Provincial AI developments include Quebec AI regulatory developments. AI-IDP's cooperative-federalism model preserves provincial authority over provincial matters while establishing a national minimum standard. Provincial monitoring is documented as a future work item.

7. Governance Gap

The governance gap is that current Canadian law and policy do not provide a universal, permanent, tamper-evident record of AI agent actions and their authority chains, applicable to all AI agents operating in Canada, with privacy safeguards, Indigenous data governance, tiered conformance, and cooperative-federalism enforcement. Existing instruments (PIPEDA, Privacy Act, Treasury Board Directive, AIA) address aspects of AI accountability but do not provide the universal identity, traceability, and audit framework that AI-IDP provides. AI-IDP fills this gap.

8. Proposed National Standard

The proposed national standard is AI-IDP — Universal AI Identity, Delegation, Provenance, Traceability, Quality, Accountability, and Permanent Audit Standard. The standard comprises: (1) 25 specification documents (spec/); (2) 14 JSON schemas (schemas/); (3) four conformance levels (L1–L4); (4) the AegisTrace reference implementation (src/aegistrace/); (5) the test suite (tests/); (6) the threat model (threat-model/); (7) the impact reports (impact/); (8) the administrative implementation plan (administration/); (9) the government package (government/); (10) the arXiv research paper (paper/).

9. Legal Objective

The legal objective is to establish, under Canadian law, that every operational AI agent used, created, deployed, distributed, controlled, executed, or made available in Canada must possess a unique persistent AI Actor Identifier, runtime instance identifiers, traceable relationships, a permanent append-only tamper-evident history, quality evidence, registry participation, and conformity certification. The intended Canadian normative principle is "no valid AI actor identity, no lawful agent operation; no valid authority chain, no lawful material action; no permanent trace record, no lawful mutating action; no verifiable user-agent-model-provider chain, no claim of accountable AI operation; no executed quality evidence, no claim that AI-generated code or systems are verified, tested, secure, compliant, or production-ready." This is the proposed legal objective. It is not current Canadian law.

10. Technical System

The technical system (AegisTrace) provides: persistent AI Actor identifiers with permanent uniqueness; runtime agent instance identifiers; provider/model/deployment/controller/principal/agent registries; delegation broker with bounded scope; policy engine with approval single-use; event collector with Ed25519 signatures and SHA-256 hash chains; append-only ledger with tamper-evidence; Merkle anchoring for public verification; filesystem, Git, GitHub, database, and MCP adapters; CLIs for verify, audit, reconstruct, and admin; FastAPI server for integration; SQLite storage with PostgreSQL-compatible production design; offline mode with reconciliation; federation protocol. The reference implementation passes 113 of 113 tests with the test-full profile, including targeted security, privacy, permanence, and conformance tests.

11. Registry Design

The registry design provides four visibility tiers: PUBLIC (public provider identities, model identities, agent identities, registry authorities, public verification keys, protocol versions, schema versions, revocation status, certification status, conformity status, signed ledger roots, Merkle roots, release attestations, public schemas, public specifications); CONTROLLED (provider registrations, model registrations, deployment records, controller records, agent definitions, agent-instance records, task/delegation/authorization records, action ledgers, resource manifests, quality evidence, incidents, revocations, corrections, audit reports — accessible to authorized regulators, auditors, certification bodies); ORGANIZATION-PRIVATE (user and principal identities — pseudonymous externally; full action ledgers; full evidence payloads; internal policy and approval records); SEALED (sealed identity resolution, sealed evidence payloads, sealed approval records — accessible only under judicial or regulator-controlled disclosure).

12. Permanent Trace Design

The permanent trace design provides: append-only events with SHA-256 hash chaining; Ed25519 signatures; Merkle anchoring to public verification surfaces; independent archival replication; corrections as new signed events (no history erasure); revocations as new signed events (history preserved); disputes as new signed events; redactions separately authorized and recorded; legal holds preserving records; sealed records for sensitive information; long-term signature migration; content separation (permanent minimal metadata and cryptographic commitments, not full content); access logging for non-public records.

13. Quality-Assurance Design

The quality-assurance design provides: every quality claim (tested, verified, implemented, validated, rendered, visually inspected, reproducible, publication-ready, audit-ready, complete, secure, compliant, production-ready) resolves to evidence (test runs, builds, releases, attestations, audits, certifications). The completion language (Master Prompt §37) is normative. AI-generated code carries provenance evidence (agent, model, version, principal, test coverage, test results, code review, static analysis, reproduction). Supply-chain evidence includes SBOM (SPDX or CycloneDX), SLSA provenance, in-toto attestations, Sigstore signatures, Rekor entries.

14. Privacy Safeguards

The privacy safeguards include: minimization (only necessary personal information collected); purpose limitation; pseudonymization (user and principal identifiers pseudonymous by default); sealed records (sensitive information sealed and access-controlled); content separation (permanent metadata separated from content); access control (access to sealed records logged and auditable); correction and revocation (corrections and revocations as new signed events); retention (permanent minimal metadata and commitments retained indefinitely; full content per retention schedule); transparency (public registry entries expose only non-sensitive fields); recourse (documented mechanism for disputes, corrections, and revocations).

15. Human-Rights Safeguards

The human-rights safeguards include: Charter compliance (ss. 2(b), 7, 8, 15 — analyzed in government/CHARTER_ANALYSIS.md); privacy-law compliance (PIPEDA, Privacy Act, provincial statutes — analyzed in government/PRIVACY_AND_HUMAN_RIGHTS_ANALYSIS.md); Indigenous data governance (OCAP® principles, First Nations, Inuit, and Métis distinctions, TRC Calls to Action — analyzed in research/synthesis/INDIGENOUS_DATA_GOVERNANCE_ANALYSIS.md); access to justice (recourse mechanism accessible without legal representation; provincial legal-aid coverage; admissibility in Canadian courts); non-discrimination (monitoring for discriminatory effects; accessible recourse; small-developer and open-source profiles reduce barriers).

16. Indigenous Data-Governance Considerations

Indigenous data-governance considerations include: recognition of Indigenous data sovereignty; OCAP®-aligned handling (Ownership, Control, Access, Possession); First Nations, Inuit, and Métis distinctions-based approach; TRC Calls to Action (particularly Calls to Action 43–44 on UNDRIP adoption and Calls to Action 7, 18, 19 on aboriginal health and reconciliation); community-controlled access for Indigenous community data; consultation with rights-holders as a precondition for implementation. The Indigenous data-governance analysis is grounded in public frameworks. Consultation with rights-holders is a precondition for implementation but not performed in this proposal.

17. Business and Economic Impact

The business and economic impact analysis (impact/business-and-operations/BUSINESS_AND_OPERATIONS_IMPACT_REPORT.md) finds that AI-IDP creates a net positive business and operational impact over a five-to-ten-year horizon for most Canadian organizations. Payback in 18–30 months for mid-size enterprises. Net positive for small developers at L1. Insurance premium reductions projected at 10–25% for L3–L4 certified operators. Average per-incident reconstruction cost reduction 70–90%.

18. HR and Labour Impact

The HR and labour impact analysis (impact/human-resources/HR_AND_LABOUR_IMPACT_REPORT.md) finds that AI-IDP can improve workplace accountability and reduce disputes over AI-related decisions, but creates risks of employee surveillance, disciplinary misuse, and erosion of worker autonomy. The privacy safeguards are essential. Collective agreements should specify permitted HR uses of AegisTrace data. The standard's success depends on equitable implementation, collective-agreement engagement, and worker recourse mechanisms that are documented, accessible, and enforceable.

19. Social Impact

The societal impact assessment (impact/societal/SOCIETAL_IMPACT_ASSESSMENT.md) finds that AI-IDP creates substantial societal benefits (improved accountability, reduced harm, faster incident response, stronger consumer protection, enhanced democratic oversight) while creating societal risks (surveillance potential, permanent-record chilling effects, market concentration, digital-divide exacerbation). The standard's success depends on robust privacy safeguards, equitable implementation, public consultation, civil-society engagement, and ongoing democratic oversight. The assessment explicitly does not frame surveillance as accountability without examining consequences.

20. Administrative Model

The administrative model (administration/ADMINISTRATIVE_IMPLEMENTATION_PLAN.md) provides: national registry authority (operator of public and controlled tiers); sectoral registry authorities (for regulated sectors); provincial registry authorities (where provinces establish equivalent regimes); certification bodies (accredited conformity assessors); auditor registry (accredited auditors); incident coordinator (national coordinator for cross-sectoral AI incidents); public-sector profile; private-sector profile; open-source profile; small-developer profile; offline profile; critical-infrastructure profile.

21. Implementation Options

The implementation options include: standards route (Standards Council of Canada — supported by research/synthesis/STANDARDS_CROSSWALK.md and the normative spec/ set); administrative route (Treasury Board Directive amendment — government/AIA_EXTENSION_PROPOSAL.md); legislative route (proposed AI Actor Identity and Traceability Act — government/PROPOSED_AI_ACTOR_IDENTITY_AND_TRACEABILITY_ACT.md); procurement route (federal procurement profile — government/FEDERAL_PROCUREMENT_PROFILE.md); AIA extension route (government/AIA_EXTENSION_PROPOSAL.md); cooperative-federalism route (national minimum standard with provincial equivalent-or-stronger regimes — government/FEDERAL_PROVINCIAL_JURISDICTION_ANALYSIS.md). The recommended approach is multi-pathway with public consultation.

22. Procurement Route

The procurement route (government/FEDERAL_PROCUREMENT_PROFILE.md) integrates AI-IDP registration into federal procurement. Federal procurement of AI-enabled systems requires AI Actor identifiers, conformance-level certifications (L3 or higher for federal procurement), evidence of permanent traceability, and integration with the procuring organization's registry. Procurement contracts specify conformance-level requirements, incident-reporting timelines, evidence-preservation periods, and audit-access terms.

23. AIA Route

The AIA route (government/AIA_EXTENSION_PROPOSAL.md) modifies the Treasury Board Algorithmic Impact Assessment to require AI-IDP registration as a field, with trace-evidence requirements for high-impact automated decisions. The extension integrates AI-IDP with the existing federal automated-decision framework without replacing it.

24. Standards Route

The standards route uses this national project proposal as the cover document, research/synthesis/STANDARDS_CROSSWALK.md as the standards-alignment record, and the spec/ set as normative technical content for a proposed Standards Council of Canada review. The standards route provides voluntary adoption with sectoral certification.

25. Legislative Route

The legislative route (government/PROPOSED_AI_ACTOR_IDENTITY_AND_TRACEABILITY_ACT.md, government/LEGISLATIVE_INSTRUCTIONS.md, government/DRAFT_STATUTORY_PROVISIONS_FOR_LEGAL_REVIEW.md, government/DRAFT_REGULATORY_FRAMEWORK.md) enacts the proposed AI Actor Identity and Traceability Act. The Act establishes the legal requirement for AI Agent registration, the registry authority, conformance levels, enforcement options, penalty options, civil-liability options, evidentiary options, and incident-reporting rules. The Act is distinguished from current law and from AIDA throughout.

26. Enforcement Options

The enforcement options (government/ENFORCEMENT_OPTIONS.md) include: civil penalties (administrative monetary penalties); criminal liability (for fraudulent attribution, evidence tampering, operating without registration); service suspension (regulator authority to suspend non-compliant services); procurement bar (non-compliant agents barred from federal procurement); civil liability (statutory civil liability for harms caused by non-compliant operation); evidentiary consequences (non-compliant operation creates evidentiary presumptions).

27. Risk Analysis

The risk analysis (project-control/RISK_REGISTER.md) identifies 15 material risks with mitigations. Key risks include: permanent identifiers becoming surveillance enablers (mitigated by pseudonymous identifiers and sealed identity resolution); cryptographic obsolescence (mitigated by interface-isolated signing and PQC migration documentation); provider foreign-hosting avoidance (mitigated by jurisdictional nexus test and controlling-organization liability); provincial jurisdiction conflict (mitigated by cooperative-federalism model); small-developer/open-source burden (mitigated by tiered conformance and compliance profiles); registry compromise (mitigated by tamper-evidence and independent replication); malicious administrator erasure (mitigated by independent archival replication and Git history rewriting detection); privacy-vs-permanence legal conflict (mitigated by encrypted evidence payloads and content separation); employee surveillance abuse (mitigated by labour-impact report and worker recourse mechanism); Indigenous data sovereignty conflict (mitigated by OCAP-aligned handling and consultation precondition); compliance market concentration effect (mitigated by open standard and small-developer profile); false confidence in AI-generated code quality (mitigated by quality-evidence protocol and explicit completion language); foreign-provider non-cooperation (mitigated by controlling-organization liability and enforcement options); scope creep (mitigated by scope-lock document and extension policy); project execution drift (mitigated by authority guardian role and decision log).

28. Pilot Proposal

The pilot proposal recommends a phased rollout: (Phase 1, year 1) federal public-sector pilot — Treasury Board, ISED, CAISI, and selected federal departments deploy AI-IDP at L3 for high-impact automated decisions; (Phase 2, year 2) regulated-sector pilot — financial institutions, healthcare agencies, and critical infrastructure operators deploy AI-IDP at L3–L4 with sectoral certification; (Phase 3, year 3) provincial pilot — selected provinces deploy AI-IDP at L2–L3 with federation to the national registry; (Phase 4, year 4) universal rollout — all AI agents operating in Canada required to register at L1 or higher, with sectoral and use-case-specific conformance-level requirements; (Phase 5, year 5) enforcement — civil penalties, procurement bar, and evidentiary consequences activated for non-compliant operation.

29. Evaluation Framework

The evaluation framework provides: annual regulator reporting to Parliament; independent audit of the national registry authority; sectoral certification body performance review; provincial federation performance review; civil-society oversight board; academic research program; public consultation every three years; legislative review every five years. The evaluation framework is documented in administration/PERFORMANCE_MEASURES.md.

30. Recommendations

The recommendations are:

- 1 Adopt AI-IDP as a National Standard of Canada through the Standards Council of Canada, with the technical specification as the normative content.
- 2 Integrate AI-IDP registration into the Treasury Board Algorithmic Impact Assessment, with trace-evidence requirements for high-impact automated decisions.
- 3 Establish tiered conformance levels (L1–L4) with sectoral certification bodies for regulated sectors (finance, healthcare, public sector, critical infrastructure).
- 4 Establish a national AI-IDP registry authority with provincial federation agreements under the cooperative-federalism model.
- 5 Establish small-developer and open-source compliance profiles with reduced-burden paths, no-fee L1 registration, and mentorship programs.
- 6 Establish enforcement options including civil penalties, service suspension, procurement bar, civil liability, and evidentiary consequences.
- 7 Enact the proposed AI Actor Identity and Traceability Act establishing the legal requirement for AI Agent registration, the registry authority, conformance levels, enforcement, and incident-reporting rules.
- 8 Establish incident-reporting rules requiring regulator notification for high-severity and critical incidents, and affected-person notification for privacy incidents.
- 9 Establish insurance and liability frameworks including AI liability insurance products and statutory civil liability for harms caused by non-compliant operation.
- 10 Conduct public consultation through a multi-channel process (online portal, in-person sessions, targeted consultation with Indigenous communities, civil-society organizations, industry, sectoral regulators, parliamentary committee hearings, Standards Council of Canada standards-development process).

31. Appendices

Appendices include: (A) full invariant list (spec/AI-IDP-CORE.md Section 5); (B) full action vocabulary (spec/RESOURCE_TRACE_PROTOCOL.md); (C) full visibility-tier definitions (spec/REGISTRY_PROTOCOL.md); (D) full conformance-level definitions (spec/CONFORMANCE_LEVELS.md); (E) full threat-model summary (threat-model/THREAT_MODEL.md); (F) full test-suite summary (tests/); (G) full cost-benefit analysis (impact/business-and-operations/COST_BENEFIT_ANALYSIS.md); (H) full jurisdiction analysis (government/FEDERAL_PROVINCIAL_JURISDICTION_ANALYSIS.md); (I) full Charter analysis (government/CHARTER_ANALYSIS.md); (J) full privacy and human-rights analysis (government/PRIVACY_AND_HUMAN_RIGHTS_ANALYSIS.md).

32. Evidence Tables

Evidence tables include: (1) the evidence register (research/registers/EVIDENCE_REGISTER.csv); (2) the contradiction register (research/contradictions/CONTRADICTION_REGISTER.md); (3) the source-gap register (project-control/SOURCE_GAP_REGISTER.md); (4) the assumption register (project-control/ASSUMPTION_REGISTER.md); (5) the decision log (project-control/DECISION_LOG.md); (6) the validation status (project-control/VALIDATION_STATUS.md).